Chapter 10: Liquids and Solids

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 10

Part I

Intermolecular Forces

Outline of Part I: Intermolecular Forces

Outline of Part I: Intermolecular Forces

Chapter 10: Liquids and Solids

Gas-phase molecules are far apart and nearly independent. In **liquids** and **solids**, molecules are close together and strongly influenced by their neighbors.

Alice Ball — A Chemist Who Changed Medicine

- Alice Ball (1892–1916) was the first chemist to isolate chaulmoogra oil's active component and develop water-soluble injectable esters.
- Her technique — the "Ball Method" — became the standard leprosy treatment for over 20 years.
- The **viscosity** (resistance to flow) of the crude oil was due to strong **intermolecular forces** — the same forces we study in this chapter.

Understanding intermolecular forces explains physical properties: boiling point, viscosity, surface tension, and the structure of solids.

Chapter Overview

Topics Covered in Chapter 10

1. **Intermolecular Forces** — dispersion, dipole-dipole, hydrogen bonding, ion-dipole
2. **Properties of Liquids** — surface tension, viscosity, capillary action
3. **Phase Transitions** — vaporization, condensation, melting, freezing, sublimation; vapor pressure; Clausius-Clapeyron equation
4. **Phase Diagrams** — triple point, critical point, water and CO_2 diagrams
5. **The Solid State** — crystalline vs. amorphous; ionic, metallic, covalent, molecular solids
6. **Lattice Structures** — unit cells, packing efficiency, density, X-ray diffraction

Outline of Part I: Intermolecular Forces

Intermolecular vs. Intramolecular Forces

Intramolecular Forces

Within a molecule — chemical bonds (covalent, ionic, metallic).

Very strong: breaking H–O bonds in water requires $\sim 925\ \frac{\text{kJ}}{\text{mol}}$.

Intermolecular Forces (IMFs)

Between molecules — electrostatic attractions between partial or full charges.

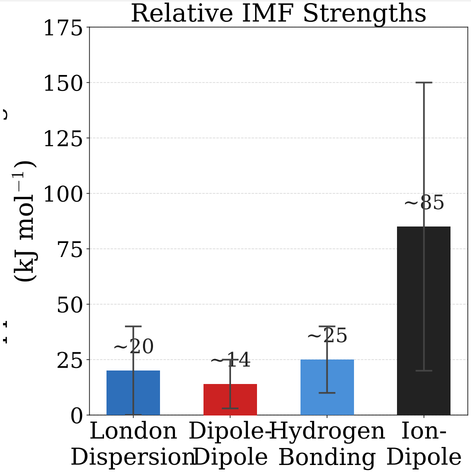
Much weaker: breaking HCl IMFs requires only $\sim 17\ \frac{\text{kJ}}{\text{mol}}$.

IMFs are about **5–25 times weaker** than covalent bonds, yet they control melting points, boiling points, viscosity, and surface tension.

Do not confuse: boiling **breaks IMFs**, not covalent bonds. Water molecules remain intact when water boils.

The Four Types of IMFs

- **London Dispersion Forces** — present in *all* substances; caused by temporary dipoles
- **Dipole-Dipole Attractions** — between polar molecules with permanent dipoles
- **Hydrogen Bonding** — extra-strong dipole-dipole for H bonded to F, O, or N
- **Ion-Dipole Forces** — between an ion and a polar molecule



Strength generally:
London < Dipole-Dipole < H-Bond <

London Dispersion Forces

Definition

London dispersion forces (also called **van der Waals forces** or **induced-dipole forces**) arise from *temporary, instantaneous* dipoles created by random fluctuations in electron distribution.

How They Arise

- At any instant, electrons may be unevenly distributed $\Rightarrow$ **temporary dipole**.
- This temporary dipole induces a dipole in a neighboring atom/molecule.
- The two dipoles attract each other.
- These attractions are short-lived but occur constantly.

London forces are present in **every** substance — atoms, nonpolar molecules, polar molecules, and ions. They are the *only* IMF in nonpolar substances.

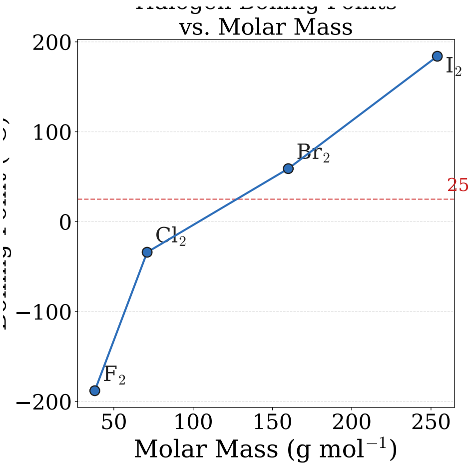
Dispersion Forces: Size and Shape

What Controls Their Strength?

- **Molar mass / number of electrons**: More electrons $\Rightarrow$ more polarizable $\Rightarrow$ stronger forces.
- **Molecular shape**: Elongated molecules have more surface contact area $\Rightarrow$ stronger forces.

Halogen Trend

Substance	MW	State @ 25°C
F_2	$38\ \frac{g}{mol}$	gas
Cl_2	$71\ \frac{g}{mol}$	gas
Br_2	$160\ \frac{g}{mol}$	liquid



Dispersion Forces: Molecular Shape Effect

Pentane Isomers

All three isomers have the same molecular formula (C_5H_{12}, MW $= 72\,\frac{\text{g}}{\text{mol}}$) but different boiling points:

Isomer	**Shape**	**Boiling Point**
n-pentane	linear (elongated)	36.1 °C
isopentane	branched	27.7 °C
neopentane	compact (spherical)	9.5 °C

More elongated ⇒ greater surface contact area ⇒ stronger dispersion forces ⇒ **higher boiling point**.
Compact/spherical shape ⇒ less contact area ⇒ weaker forces ⇒ **lower boiling point**.

Check Your Understanding

Which substance has the **stronger** London dispersion forces?

(a) CH_4 or SiH_4?

(b) n-pentane or neopentane?

(c) Br_2 or Cl_2?

Come to lecture to see the answer.

Dipole-Dipole Attractions

Definition

Dipole-dipole attractions are electrostatic forces between the partially positive (δ^+) end of one polar molecule and the partially negative (δ^-) end of a neighboring polar molecule.

Example: HCl

HCl is polar ($\mu \neq 0$); the δ^+ H end of one HCl aligns with the δ^- Cl end of another.

HCl boils at $-85\,°C$; nonpolar Ar ($40\,\frac{g}{mol}$, similar MW) boils at $-186\,°C$.

The dipole-dipole attraction raises the boiling point significantly.

Dipole-dipole forces only operate when molecules are **aligned**.

In solids, molecules are well-aligned.
In liquids, alignment is partial and temporary.

Hydrogen Bonding

Definition

Hydrogen bonding is an especially strong dipole-dipole interaction that occurs when hydrogen is bonded to a highly electronegative atom — **F**, **O**, or **N** ("FON").

Why Is It So Strong?

- H is tiny; when bonded to F, O, or N, the shared electrons are pulled far away.
- H is nearly a "bare proton" — its δ^+ charge is concentrated in a very small volume.
- The result is a very strong partial charge that attracts the lone pairs of F, O, or N on neighboring molecules.

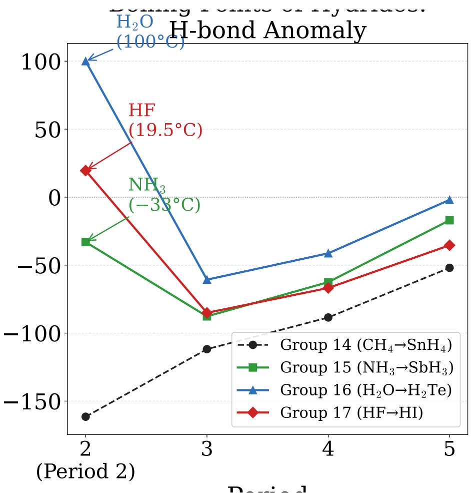
H-bonds are $\sim$5–10% as strong as covalent bonds, but **much stronger** than ordinary dipole-dipole forces. They occur between **different** molecules

Hydrogen Bonding: The Anomaly of H_2O, HF, NH_3

For group 14 hydrides (CH_4, SiH_4, GeH_4, SnH_4), boiling points increase smoothly with molar mass — as expected for dispersion forces only.

But for groups 15, 16, and 17, the lightest hydrides (**NH_3, H_2O, HF**) have **anomalously high** boiling points because of hydrogen bonding.

Without H-bonding, water would boil near $-70\,°C$; life as we know it would be impossible.



Real-World Examples of IMFs

Gecko Adhesion — Dispersion Forces

Gecko feet have millions of tiny hairs (setae) that branch into nanoscale tips.

The enormous surface area between tips and the wall creates enough **London dispersion forces** to support many times the gecko's weight.

DNA — Hydrogen Bonding

Complementary base pairs are held together by **hydrogen bonds**:

- Adenine-Thymine: **2** H-bonds
- Cytosine-Guanine: **3** H-bonds

These bonds are weak enough that DNA can "unzip" for replication, yet strong enough to maintain the double helix.

IMFs may be weak individually, but they are enormously powerful **collectively**.

Ion-Dipole Forces

Definition

Ion-dipole forces are attractions between an **ion** (cation or anion) and the partially charged end of a **polar molecule**.

Example: NaCl Dissolving in Water

- Na^+ ions attract the δ^- oxygen of H_2O (water's O faces the cation).
- Cl^- ions attract the δ^+ hydrogen atoms of H_2O.
- Each ion is surrounded by a **hydration shell** of water molecules.
- Ion-dipole forces are strong enough to pull ions out of the crystal lattice.

Ion-dipole forces are the **strongest** of the four IMF types. They are responsible for the solubility of ionic compounds in polar solvents like water.

Check Your Understanding

Identify the **dominant** IMF(s) in each substance:

(a) I_2 (nonpolar, solid at room temp)

(b) HCl (polar)

(c) CH_3OH (methanol, has O–H bond)

(d) NaBr dissolved in water

Come to lecture to see the answer.

Part II

Properties of Liquids

Outline of Part II: Properties of Liquids

Properties of Liquids

Outline of Part II: Properties of Liquids

Properties of Liquids

Surface Tension

Definition

Surface tension is the energy required to increase the surface area of a liquid (units: $\frac{\text{J}}{\text{m}^2}$ or $\frac{\text{N}}{\text{m}}$).

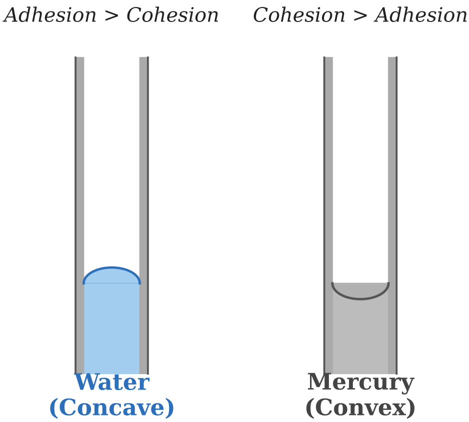
Why Does It Occur?

Molecules *at* the surface are only attracted by molecules *below* and *beside* them — the net force pulls them inward.

Molecules in the *interior* are attracted from all directions.

Liquids minimize their surface area $\Rightarrow$ water drops are spherical.

Meniscus: Water vs. Mercury



Cohesion, Adhesion, and the Meniscus

Cohesive Forces

IMFs between **identical** molecules (liquid–liquid).

Example: water molecules attract each other via H-bonds.

Adhesive Forces

IMFs between **different** molecules (liquid–surface).

Example: water attracts glass (SiO_2) via H-bonds.

The Meniscus

- **Water in glass**: adhesive > cohesive ⇒ **concave** meniscus (wets glass).
- **Mercury in glass**: cohesive > adhesive ⇒ **convex** meniscus (does not wet glass).

Always read a liquid volume at the **bottom** of a concave meniscus.

Viscosity

Definition

Viscosity is a liquid's resistance to flow. Units: pascal-second (Pa s) or millipascal-second (mPa s).

Substance	**Viscosity (mPa·s, 25°C)**	**IMF**
Octane	0.508	London (nonpolar)
Water	0.890	H-bonding
Ethanol	1.074	H-bonding
Ethylene glycol	16.1	Many H-bonds
Honey	$\sim$2000–10000	Many H-bonds + size

Stronger IMFs $\Rightarrow$ **higher viscosity**.
Higher temperature $\Rightarrow$ more kinetic energy $\Rightarrow$ **lower viscosity** (molecules can

Check Your Understanding

(a) Ethylene glycol (antifreeze) has two –OH groups per molecule; ethanol has one. Which is more viscous? Why?

(b) Motor oil is more viscous at 0 °C than at 100 °C. Is this consistent with your knowledge of IMFs and temperature?

Come to lecture to see the answer.

Capillary Action

Definition

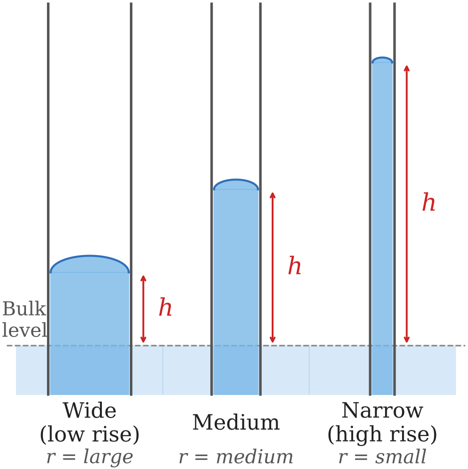
Capillary action is the spontaneous rise (or fall) of a liquid in a narrow tube due to the interplay of adhesive and cohesive forces.

Capillary Rise Formula

$$h = \frac{2T\cos\theta}{r\rho g}$$

- h = height of rise
- T = surface tension
- θ = contact angle (water/glass: $\approx 0°$)

Capillary Rise: $h = \frac{2T\cos\theta}{r\rho g}$



Example: Capillary Rise

Water in a Glass Capillary — (chem2e ch10)

Water rises in a glass capillary tube with radius 0.25 mm at 25 °C.
Given: $T = 71.99\,\frac{\mathrm{mN}}{\mathrm{m}}$, $\theta \approx 0°$, $\rho = 997\,\frac{\mathrm{kg}}{\mathrm{m}^3}$, $g = 9.8\,\frac{\mathrm{m}}{\mathrm{s}^2}$.
How high does the water rise?

$$h = \frac{2\,T\cos\theta}{r\rho g}$$

$$h = \frac{2 \times 0.071\,99\,\frac{\mathrm{N}}{\mathrm{m}} \times \cos(0°)}{2.5 \times 10^{-4}\,\mathrm{m} \times 997\,\frac{\mathrm{kg}}{\mathrm{m}^3} \times 9.8\,\frac{\mathrm{m}}{\mathrm{s}^2}}$$

$$\boxed{h \approx 0.059\,\mathrm{m} = 5.9\,\mathrm{cm}}$$

$\cos(0°) = 1$ (water fully wets glass).

Narrowing the tube by 10× would raise water 10× higher.

Part III

Phase Transitions

Outline of Part III: Phase Transitions

Phase Transitions

Outline of Part III: Phase Transitions

Phase Transitions

Overview: Phase Transitions

A **phase transition** is a change from one state of matter to another.

Transition	From	To	Energy
Melting (fusion)	solid	liquid	endothermic
Freezing	liquid	solid	exothermic
Vaporization	liquid	gas	endothermic
Condensation	gas	liquid	exothermic
Sublimation	solid	gas	endothermic
Deposition	gas	solid	exothermic

Energy is required to **separate** molecules (endothermic = breaking IMFs).
Energy is **released** when molecules come together (exothermic = forming IMFs).

Vapor Pressure

Definition

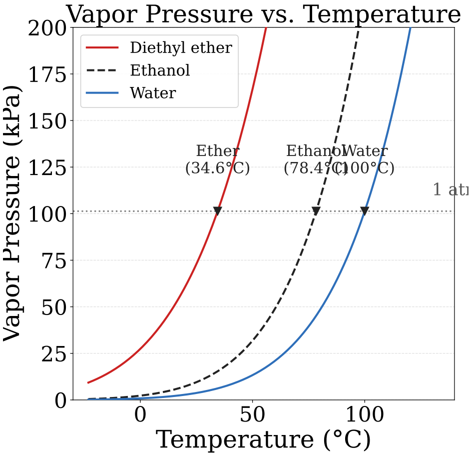
Vapor pressure is the pressure exerted by a vapor in equilibrium with its liquid (or solid) in a closed container at a given temperature.

Dynamic Equilibrium

In a closed container, vaporization and condensation eventually reach the **same rate**:

$$\text{liquid} \rightleftharpoons \text{gas}$$

The vapor pressure depends on **temperature** and the **strength of IMFs**.



Boiling Point

Definition

Boiling occurs when a liquid's vapor pressure equals the external (atmospheric) pressure. Bubbles of vapor form throughout the liquid.

Normal Boiling Point

The **normal boiling point** is the temperature at which vapor pressure equals exactly 101.3 kPa (1 atm).

- At **high altitude** (lower P_{atm}), water boils below 100 °C.
  e.g. at Denver (1.6 km), water boils at ≈ 94 °C — eggs take longer to cook.
- In a **pressure cooker** (higher P), water boils above 100 °C — food cooks faster.

Boiling is **not** the same as evaporation. Evaporation occurs at any temperature

The Clausius-Clapeyron Equation

Vapor Pressure and Temperature

Vapor pressure increases **exponentially** with temperature:

$$\ln P = -\frac{\Delta H_{\text{vap}}}{R} \cdot \frac{1}{T} + \ln A$$

Two-Point Form — **Most Useful for Calculations**

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

- $P_1,\ P_2$ = vapor pressures at temperatures $T_1,\ T_2$ (in K)
- ΔH_{vap} = enthalpy of vaporization $\left(\frac{\text{J}}{\text{mol}}\right)$
- $R = 8.314\,\frac{\text{J}}{\text{mol K}}$

Example: Clausius-Clapeyron Equation

Vapor Pressure of Water — (chem2e ch10)

The vapor pressure of water is 1.94 kPa at 17 °C and 3.61 kPa at 27 °C. Calculate ΔH_{vap} for water.

$$T_1 = 290\,\mathrm{K},\ T_2 = 300\,\mathrm{K}$$

$$\ln\left(\frac{3.61}{1.94}\right) = \frac{\Delta H_{\text{vap}}}{8.314\,\frac{\mathrm{J}}{\mathrm{mol\,K}}}\left(\frac{1}{290} - \frac{1}{300}\right)$$

$$0.620 = \frac{\Delta H_{\text{vap}}}{8.314} \times 1.149 \times 10^{-4}$$

$$\boxed{\Delta H_{\text{vap}} \approx 4.48 \times 10^{4}\,\frac{\mathrm{J}}{\mathrm{mol}} = 44.8\,\frac{\mathrm{kJ}}{\mathrm{mol}}}$$

Literature value: $44.0\,\frac{\mathrm{kJ}}{\mathrm{mol}}$ at 25 °C.

Good agreement! Small temperature range limits precision.

Check Your Understanding

Which of the following has the **highest** vapor pressure at room temperature?

(a) Diethyl ether (weak IMFs, low BP)
(b) Ethanol (H-bonding, moderate BP)
(c) Water (strong H-bonding, BP = 100°C)

Which one is most **volatile** (evaporates fastest)?

Come to lecture to see the answer.

Enthalpy of Phase Transitions

Quantity	**Symbol**	**Value for Water**
Enthalpy of vaporization	ΔH_{vap}	$40.67\,\frac{\text{kJ}}{\text{mol}}$ at $100\,^{\circ}\text{C}$
Enthalpy of fusion	ΔH_{fus}	$6.01\,\frac{\text{kJ}}{\text{mol}}$ at $0\,^{\circ}\text{C}$
Enthalpy of sublimation	ΔH_{sub}	$51.1\,\frac{\text{kJ}}{\text{mol}}$ (ice at $-10°\text{C}$)

$\Delta H_{\text{sub}} \approx \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$
(sublimation = melting + vaporization in sequence)

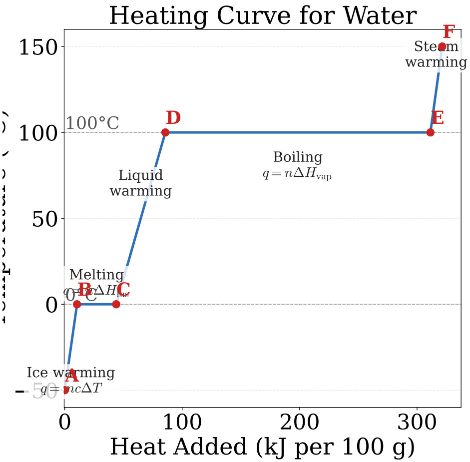
Practical Application: Perspiration

The body absorbs 40.67 kJ of heat for every mole (18 g) of sweat that evaporates from the skin — a very effective cooling mechanism.

Heating Curves

For $1\,\mathrm{g}$ of Water, Ice $\rightarrow$ Steam

- **Segment A–B**: ice warming; $q = mc\Delta T$, $c_{\mathrm{ice}} = 2.09\,\frac{\mathrm{J}}{\mathrm{g\,K}}$
- **B–C**: melting (plateau at $0\,^\circ\mathrm{C}$); $q = n\Delta H_{\mathrm{fus}}$
- **C–D**: liquid water warming; $c_{\mathrm{liq}} = 4.18\,\frac{\mathrm{J}}{\mathrm{g\,K}}$
- **D–E**: boiling (plateau at $100\,^\circ\mathrm{C}$); $q = n\Delta H_{\mathrm{vap}}$
- **E–F**: steam warming; $c_{\mathrm{gas}} = 2.01\,\frac{\mathrm{J}}{\mathrm{g\,K}}$



During phase transitions (plateaus),

Example: Heating Curve Calculation

Energy to Convert Ice to Steam

How much heat (in kJ) is required to convert 100 g of ice at −20 °C to steam at 120 °C?
Use: $\Delta H_{\text{fus}} = 6.01\,\frac{\text{kJ}}{\text{mol}}$; $\Delta H_{\text{vap}} = 40.67\,\frac{\text{kJ}}{\text{mol}}$; $MW(\text{H}_2\text{O}) = 18.0\,\frac{\text{g}}{\text{mol}}$.

$$
\begin{aligned}
q_1 &= mc\Delta T = 100\,\text{g} \times 2.09\,\frac{\text{J}}{\text{g K}} \times 20\,\text{K} = 4.18\,\text{kJ} \\
q_2 &= n\Delta H_{\text{fus}} = \frac{100}{18.0}\,\text{mol} \times 6.01\,\frac{\text{kJ}}{\text{mol}} = 33.4\,\text{kJ} \\
q_3 &= mc\Delta T = 100\,\text{g} \times 4.18\,\frac{\text{J}}{\text{g K}} \times 100\,\text{K} = 41.8\,\text{kJ} \\
q_4 &= n\Delta H_{\text{vap}} = \frac{100}{18.0}\,\text{mol} \times 40.67\,\frac{\text{kJ}}{\text{mol}} = 226\,\text{kJ} \\
q_5 &= mc\Delta T = 100\,\text{g} \times 2.01\,\frac{\text{J}}{\text{g K}} \times 20\,\text{K} = 4.02\,\text{kJ}
\end{aligned}
$$

$$\boxed{q_{\text{total}} = 4.18 + 33.4 + 41.8 + 226 + 4.02 = 309\,\text{kJ}}$$

Check Your Understanding

Looking at the heating curve for water:

(a) Why does the temperature remain constant during melting and boiling?

(b) The boiling plateau is much longer than the melting plateau. What does this tell you about ΔH_{vap} vs. ΔH_{fus}?

Come to lecture to see the answer.

Sublimation and Deposition

Definitions

- **Sublimation**: direct transition from **solid** → **gas** (endothermic).
- **Deposition**: direct transition from **gas** → **solid** (exothermic).

Dry Ice (CO_2)

CO_2 sublimes at $-78.5\,°C$ at 1 atm (no liquid phase at normal pressure).

Used in fog machines, food preservation, and refrigeration.

Frost / Freeze-Drying

Frost forms by deposition of water vapor directly onto cold surfaces.

Freeze-drying preserves food by sublimating ice at low pressure — no liquid water ⇒ no structural damage to food.

Part IV

Phase Diagrams

Outline of Part IV: Phase Diagrams

Outline of Part IV: Phase Diagrams

Phase Diagrams

Phase Diagrams

Definition

A **phase diagram** is a graph of pressure (y-axis) vs. temperature (x-axis) that shows the physical state of a substance under given conditions and the boundaries between phases.

Key Features

- **Regions**: areas where solid, liquid, or gas is the stable phase.
- **Curves**: boundaries where two phases coexist in equilibrium.
- **Triple point**: the unique T and P where all three phases coexist.
- **Critical point**: the T and P above which distinct liquid and gas phases cannot exist; above this point a **supercritical fluid** forms.

Phase Diagram: Water

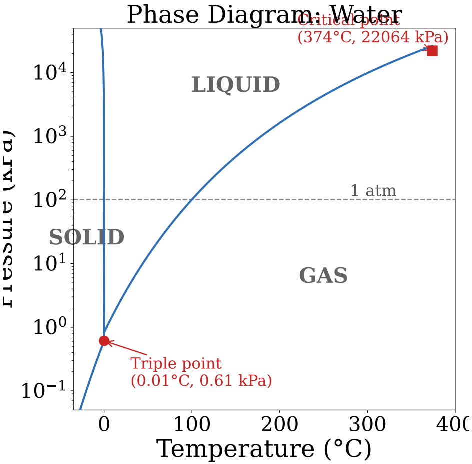
Key Data for H_2O

- **Triple point**: 0.01 °C, 0.611 kPa
- **Critical point**: 374 °C, 22.064 kPa
- **Normal melting point**: 0 °C at 101.3 kPa
- **Normal boiling point**: 100 °C at 101.3 kPa

Unusual Feature

Water's **solid-liquid boundary slopes to the left** (negative slope): increasing pressure *lowers* the melting point.

Reason: ice is *less dense* than liquid water



Phase Diagram: Carbon Dioxide

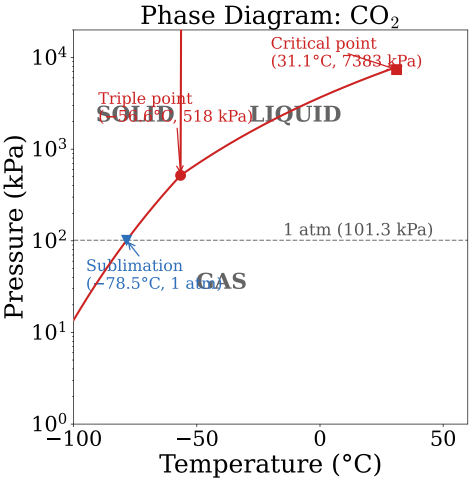
Key Data for CO_2

- **Triple point**: $-56.6\,°C$, $517\,kPa$ ($5.11\,atm$)
- **Critical point**: $31.1\,°C$, $7383\,kPa$ ($72.8\,atm$)
- At $1\,atm$, CO_2 **sublimes** at $-78.5\,°C$

Why Dry Ice Sublimes

The triple point of CO_2 lies *above* $1\,atm$. At atmospheric pressure, liquid CO_2 cannot exist — CO_2 goes directly from solid to gas.

Solid-liquid boundary: positive slope



Supercritical Fluids

Definition

Above the **critical temperature** and **critical pressure**, a substance exists as a **supercritical fluid** — with properties intermediate between liquid and gas:

Like a Liquid

- High density
- Can dissolve substances

Like a Gas

- Low viscosity
- Low surface tension
- Diffuses rapidly

Supercritical CO_2 Applications

- **Decaffeination**: sc-CO_2 selectively dissolves and removes caffeine from coffee beans.
- **Dry cleaning**: non-toxic alternative to perchloroethylene

Check Your Understanding

Using the water phase diagram:

(a) What happens to ice if pressure is increased at $-2\,^{\circ}\mathrm{C}$?

(b) Starting at $-30\,^{\circ}\mathrm{C}$ and $0.01\,\mathrm{kPa}$, what happens as temperature increases at constant pressure?

(c) Why do ice skates melt ice under the blade?

Come to lecture to see the answer.

Part V

Solids

Outline of Part V: Solids

Outline of Part V: Solids

Crystalline vs. Amorphous Solids

Crystalline Solids

Atoms, ions, or molecules arranged in a **regular, repeating pattern** (a lattice).

Properties:

- Sharp, precise melting point
- Anisotropic (properties vary by direction)

Examples: NaCl, diamond, ice, quartz, iron.

Amorphous Solids

Particles arranged **randomly**, without long-range order.

Properties:

- Soften over a temperature *range* (no sharp MP)
- Isotropic (properties same in all directions)

Examples: glass, rubber, candle wax, many plastics.

Amorphous solids are sometimes described as **supercooled liquids** — their disordered structure is “frozen in” before crystallization can occur.

Types of Crystalline Solids

Type	Particles	Bonding	Properties
Ionic	cations & anions	electrostatic	hard, brittle; high MP; nonconductor (solid)
Metallic	metal atoms + “sea” of e^-	metallic	malleable, ductile; conductive; variable MP
Covalent network	atoms	covalent bonds throughout	very hard; very high MP; usually nonconductor
Molecular	molecules	IMFs	soft; low MP; nonconductor

Ionic Solids

- Composed of **cations** and **anions** held together by **electrostatic (ionic) attraction**.
- Typically **hard** and **brittle** — shifting layers brings like charges together ⇒ repulsion ⇒ shattering.
- **High melting points** (NaCl: 801 °C; MgO: 2852 °C).
- **Non-conductive** as solid (ions fixed in place); **conductive** when molten or dissolved.

Examples

NaCl (table salt), KBr, NiO, Al_2O_3 (corundum/sapphire/ruby)

Higher ion charges ⇒ stronger attractions ⇒ higher MP.
MgO (2^+ / 2^-) melts at 2852 °C; NaCl (1^+ / 1^-) melts at 801 °C

Metallic Solids

- Composed of **metal atoms** surrounded by a delocalized "**sea of electrons**".
- Electrons flow freely $\Rightarrow$ **excellent electrical and thermal conductors**.
- Layers of atoms slide past each other $\Rightarrow$ **malleable** and **ductile**.
- **Metallic luster** due to interaction of light with free electrons.
- Melting points range widely: Hg ($-39\,°C$) to W ($3422\,°C$).

Examples

Cu, Al, Fe, Au, Na, W, Hg

Alloys are metallic solids made of two or more metals (e.g. bronze = Cu + Sn; steel = Fe + C). They often have superior properties to the pure metals.

Covalent Network Solids

- Atoms are connected by a **continuous network of covalent bonds** in 2D or 3D.
- Extremely **hard** and **high-melting** (covalent bonds must be broken throughout).
- Usually **non-conductive** (no free electrons or ions).

Diamond

Each C atom is tetrahedrally bonded to 4 others in 3D.
MP $> 3500\,^{\circ}$C. Hardest natural substance.

Graphite (Exception)

C atoms form 2D layers; layers held by weak London forces.
Soft (layers slide) and **conductive** (delocalized π electrons within layers).

Other examples: SiO_2 (quartz/sand), SiC (carborundum), Si (semiconductors).

Molecular Solids

- Composed of **neutral molecules** held by **IMFs** (London forces, dipole-dipole, H-bonds).
- **Soft** and **low-melting**: IMFs are weak compared to ionic or covalent bonds.
- **Non-conductive** (no free charges).
- Melting point increases with stronger IMFs.

Substance	**MP (°C)**	**Dominant IMF**
H_2	-259	London (nonpolar, tiny)
CO_2	-78.5 (sublimes)	London (nonpolar)
I_2	114	London (nonpolar, large)
H_2O (ice)	0	Hydrogen bonding
Table sugar	185	H-bonding + London

Check Your Understanding

Classify each solid and predict whether it has a high or low melting point:

(a) Silicon dioxide (SiO_2, quartz)

(b) Ibuprofen ($C_{13}H_{18}O_2$, a drug molecule)

(c) Potassium iodide (KI)

(d) Copper (Cu)

Come to lecture to see the answer.

Outline of Part V: Solids

The Solid State of Matter

Lattice Structures in Crystalline Solids

Unit Cells

Definition

A **unit cell** is the smallest repeating unit of a crystal lattice. Stacking unit cells in three dimensions reproduces the entire crystal.

Think of it like a single tile that, when repeated, tiles an infinite floor.

The unit cell is defined by:

- Three **edge lengths** (a, b, c)
- Three **angles** (α, β, γ) between edges

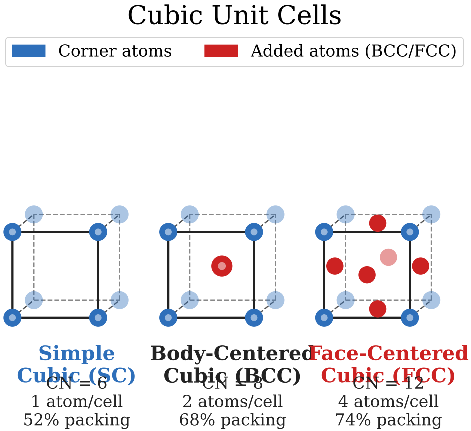
For **cubic** unit cells, $a = b = c$ and $\alpha = \beta = \gamma = 90°$.
Three types of cubic cells: **simple cubic**, **body-centered cubic**, **face-centered cubic**.

Cubic Unit Cells

	SC	BCC	FCC
Atoms in cell	1	2	4
Coord. number	6	8	12
Packing	52%	68%	74%

Examples

- SC: Polonium (only element)
- BCC: Fe, Cr, Mo, W, K, Ba
- FCC: Al, Cu, Ag, Au, Ni, Pb



The **coordination number** is the number

Atom Counting in Unit Cells

Fraction of Atom Belonging to a Unit Cell

- **Corner atom**: shared by 8 unit cells $\Rightarrow \frac{1}{8}$ per cell
- **Face atom**: shared by 2 unit cells $\Rightarrow \frac{1}{2}$ per cell
- **Edge atom**: shared by 4 unit cells $\Rightarrow \frac{1}{4}$ per cell
- **Body-center atom**: inside 1 cell $\Rightarrow$ 1 per cell

Atom Count Verification

- SC: $8 \times \frac{1}{8} = \mathbf{1}$ atom/cell
- BCC: $8 \times \frac{1}{8} + 1 = \mathbf{2}$ atoms/cell
- FCC: $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = \mathbf{4}$ atoms/cell

Example: Density from Unit Cell

Density of Aluminum — (chem2e ch10)

Aluminum forms an FCC unit cell with an edge length of 404.95 pm. Calculate the density of aluminum.
(MW of Al: 26.98 $\frac{\text{g}}{\text{mol}}$; $N_A = 6.022 \times 10^{23}\ \frac{1}{\text{mol}}$)

FCC: 4 atoms/unit cell.

$$m_{\text{cell}} = \frac{4\,\text{atoms}}{\text{cell}} \times \frac{26.98\,\text{g}}{6.022 \times 10^{23}\,\text{atoms}}$$

$$m_{\text{cell}} = 1.792 \times 10^{-22}\,\text{g}$$

$$a = 404.95\,\text{pm} = 4.0495 \times 10^{-8}\,\text{cm}$$

$$V_{\text{cell}} = a^3 = 6.647 \times 10^{-23}\,\text{cm}^3$$

$$\boxed{d = \frac{m}{V} = \frac{1.792 \times 10^{-22}\,\text{g}}{6.647 \times 10^{-23}\,\text{cm}^3} = 2.697\ \frac{\text{g}}{\text{cm}^3}}$$

Literature value for Al:
2.699 $\frac{\text{g}}{\text{cm}^3}$

Excellent agreement — unit cell geometry encodes bulk density!

Check Your Understanding

Iron (Fe) forms a BCC unit cell with an edge length of 287 pm.

(a) How many Fe atoms are in one unit cell?

(b) Is the coordination number 6, 8, or 12?

(c) Without calculating, predict whether Fe's density will be greater or less than Al's ($2.70\,\frac{\text{g}}{\text{cm}^3}$). Why?

Come to lecture to see the answer.

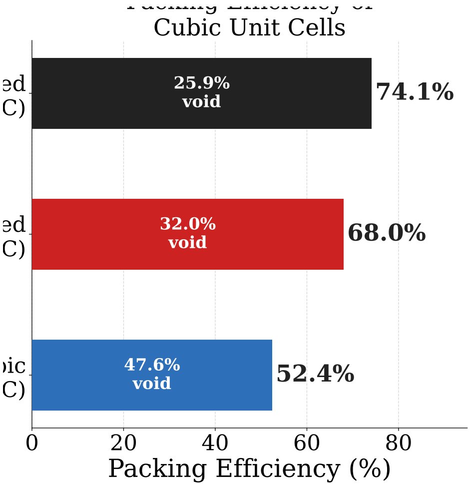
Packing Efficiency

Definition

Packing efficiency = fraction of unit cell volume occupied by atoms.

Calculated Values

- **Simple cubic:** 52% (least efficient; only Po)
- **Body-centered cubic:** 68% (common for metals)
- **Face-centered cubic:** 74% (most efficient; = hexagonal closest packing)



FCC/HCP are the most efficient ways to

Ionic Crystal Structures

Ionic crystals arrange cations into the **holes** of the anion lattice (or vice versa). The type of hole depends on the **cation-to-anion radius ratio**.

Structure	**Ratio**	**CN**	**Example**
NaCl (rock salt)	FCC anions; cations in octahedral holes	6:6	NaCl, KBr, MgO
CsCl	SC anions; cation in cubic hole	8:8	CsCl
ZnS (zinc blende)	FCC anions; cations in ½ tetrahedral	4:4	ZnS, GaAs

X-Ray Diffraction and Bragg's Law

How Crystal Structures Are Determined

X-rays are diffracted by the regularly spaced planes of atoms in a crystal. Constructive interference occurs only when Bragg's condition is met:

$$n\lambda = 2d\sin\theta$$

- $n =$ order of diffraction (integer)
- $\lambda =$ X-ray wavelength
- $d =$ spacing between crystal planes
- $\theta =$ diffraction ("glancing") angle

Example

X-rays ($\lambda = 154\,\mathrm{pm}$) diffract at $\theta = 14.2^\circ$ from a crystal plane.

Check Your Understanding

A sodium chloride (NaCl) crystal is irradiated with X-rays of wavelength 154 pm.

The first-order diffraction peak appears at $\theta = 13.7°$.

What is the spacing d between crystal planes?

Come to lecture to see the answer.

Summary: Chapter 10

- **IMFs** (London, dipole-dipole, H-bonding, ion-dipole) determine physical properties of liquids and solids.
- **Surface tension**, **viscosity**, and **capillary action** arise from the interplay of cohesive and adhesive forces.
- **Vapor pressure** increases with temperature; the Clausius-Clapeyron equation quantifies this: $\ln(P_2/P_1) = (\Delta H_{\text{vap}}/R)(1/T_1 - 1/T_2)$.
- **Heating curves** show temperature vs. heat added; plateaus occur during phase transitions.
- **Phase diagrams** map stable phases vs. P and T; triple points and critical points define special states.
- Crystalline solids are classified as **ionic, metallic, covalent network**, or **molecular** based on their bonding.
- **Cubic unit cells** (SC, BCC, FCC) are characterized by atoms/cell, coordination number, and packing efficiency; density can be calculated from